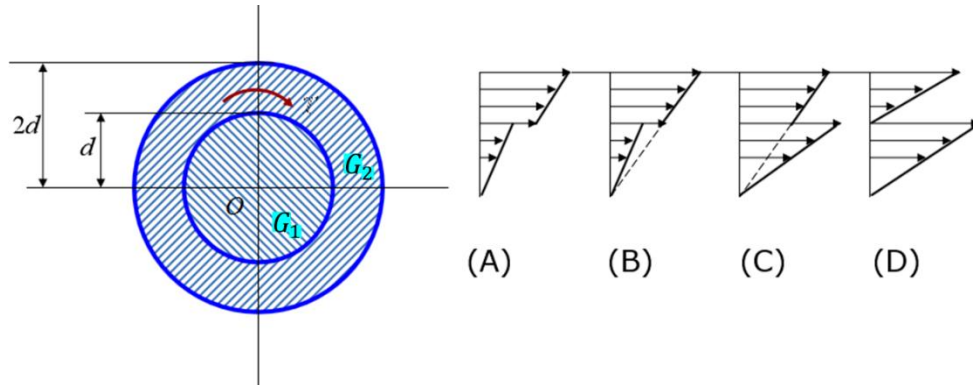


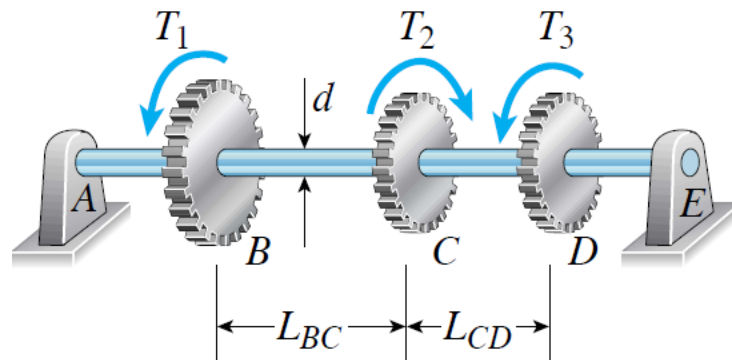
Submission: Please submit the scanned/typed document to Canvas.

1. A tube made of two different materials. The shear modulus of the outer and inner one are G_1 and G_2 , respectively. And $G_1 = 2G_2$. When a torsional loading is applied, there is no relative sliding. For the shear stress distribution on the cross-section, which one is correct?

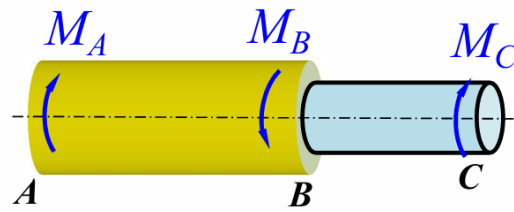


2. Figure shows a solid steel transmission shaft with a diameter $d = 30$ mm. The driving gear is C, passive gears are B and D. $T_1 = 275$ N·m, $T_2 = 450$ N·m, $T_3 = 175$ N·m. $L_{BC} = 500$ mm, $L_{CD} = 400$ mm. Shear modulus $G = 80$ GPa. Determine

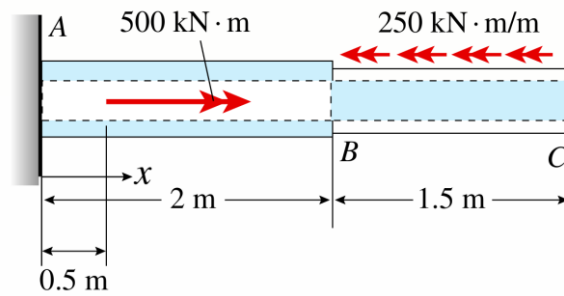
- (a) the $\tau_{\max}$ in each part of the shaft
- (b) angle of twist between gears B and D



3. Given: AB segment diameter $d_1 = 120$ mm, BC segment diameter $d_2 = 100$ mm. $M_A = 22$ kN·m, $M_B = 36$ kN·m, $M_C = 14$ kN·m. The allowable stress $[\tau] = 80$ MPa. Determine if the structure is safe.

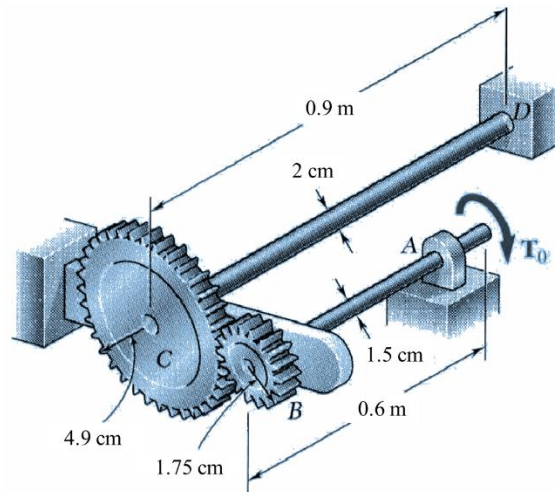


4. Two sections of steel drill pipe, joined by bolted flange plates at B, are being tested to assess the adequacy of both the pipes. In the test, the pipe structure is fixed at A, a concentrated torque of $500 \text{ kN}\cdot\text{m}$ is applied at $x = 0.5 \text{ m}$, and uniformly distributed torque intensity $t_0 = 250 \text{ kN}\cdot\text{m/m}$ is applied on pipe BC. Both pipes have the same inner diameter $d = 200 \text{ mm}$. Pipe AB has thickness $t_{AB} = 15 \text{ mm}$, while pipe BC has thickness $t_{BC} = 12 \text{ mm}$. Find the maximum shear stress and maximum twist of the pipe and their locations along the pipe. Assume $G = 75 \text{ GPa}$.



5. Two solid steel shafts are connected by gears. Knowing that for each shaft $G = 75$ GPa, and the allowable shear stress $[\tau] = 100$ MPa. Other geometric information is given in figure. Determine

- (a) The largest torque T_0 that may be applied to the end of shaft AB
- (b) The corresponding angle of twist ϕ_A at end A.



6. A circular tube of inner radius $r_1 = 3 \text{ cm}$ and outer radius $r_2 = 4.5 \text{ cm}$ is subjected to a torque produced by forces P . The forces have their lines of action at a distance $b = 10 \text{ cm}$ outside of the tube. A torsional spring of stiffness $k_R = 100 \text{ kN} \cdot \text{m}/\text{rad}$ is added at the end of the tube to resist the twist. What is the maximum value of forces P if the allowable shear stress is $[\tau] = 100 \text{ MPa}$. Assume that the tube has a length of $L = 0.5 \text{ m}$ and shear modulus $G = 70 \text{ GPa}$.

